Participant_ID	Participant_Name	Experience_Level	Mode_Order	LLM	Task_Group	Experince_Copilot (1-5)	Intro_and_Setup_min	Task1_Mode
P1	***** *****	Junior	Vibe → Agent	Claude Sonnet 4.5	Task1_Data	3	14	Vibe/Ask
P2	********* **********	Junior	Agent → Vibe	Claude Sonnet 4.5	Task1_Data	2	12	Agentic/Agent
P3	***** ******	Medium	Vibe → Agent	Claude Sonnet 4.5	Task1_Data	2	15	Vibe/Ask
P4	**** *******	Medium	Agent → Vibe	Claude Sonnet 4.5	Task1_Data	2	5	Agentic/Agent
P5	*** *******	Senior	Vibe → Agent	Claude Sonnet 4.5	Task1_Data	2	14	Vibe/Ask
P6	***** ******	Senior	Agent → Vibe	Claude Sonnet 4.5	Task1_Data	3	7	Agentic/Agent
P7	**** ******	Junior	Vibe → Agent	Claude Sonnet 4.5	Task2_Bug_Fix	1	13	Vibe/Ask
P8	*** *****	Junior	Agent → Vibe	Claude Sonnet 4.5	Task2_Bug_Fix	3	6	Agentic/Agent
P9	********* ********	Medium	Vibe → Agent	Claude Sonnet 4.5	Task2_Bug_Fix	5	5	Vibe/Ask
P10	**** *** *******	Medium	Agent → Vibe	Claude Sonnet 4.5	Task2_Bug_Fix	5	6	Agentic/Agent
P11	**** *****	Senior	Vibe → Agent	Claude Sonnet 4.5	Task2_Bug_Fix	5	9	Vibe/Ask
P12	******** ****	Senior	Agent → Vibe	Claude Sonnet 4.5	Task2_Bug_Fix	3	12	Agentic/Agent

Task1_Start_time	Task1_End_time	Task1_Duration_min	Task1_Success_0_fail_1_passed	Task1_Prompts_or_AgentRuns	Task1_Interruptions
12:32:00 PM	12:47:00 PM	15	0	4	2
1:11:00 PM	1:22:00 PM	11	0	3	1
4:13:00 PM	4:28:00 PM	15	0	8	1
2:10:00 PM	2:12:00 PM	2	1	1	0
3:12:00 PM	3:26:00 PM	14	1	4	1
9:10:00 PM	9:22:00 PM	12	0	3	0
6:18:00 PM	6:33:00 PM	15	0	2	0
8:13:00 PM	8:28:00 PM	15	0	3	0
7:06:00 PM	7:12:00 PM	6	1	5	0
7:07:00 PM	7:10:00 PM	3	1	1	0
7:18:00 PM	7:26:00 PM	8	0	5	1
10:11:00 AM	10:18:00 AM	7,2	1	1	0

Task1_Notes
P1 was unable to complete the initial project setup within the allocated time and did not create a virtual environment. Although the README explicitly described the required setup steps, Ask mode
P2 carefully read the README and inspected the files referenced within it before attempting to solve the task. Rather than directing the agent to the README, P2 manually extracted and provided
P3 carefully read the README, verified the required Python version, and initially encountered a “file not found” error due to operating in an incorrect directory. P3asked the agent to confirm screen
P4 began by reading the README and then provided both the README and task question to Agent mode. The agent handled the task end-to-end by creating the virtual environment, installing py
P5 began by carefully reading the README and ensured that all required setup steps were followed. P5 created the virtual environment but temporarily interrupted the setup to ask Ask mode for t
P6 began by reading the README in detail and created the virtual environment. As no requirements file was present, P6 manually inspected the project structure and data files, spending approxin
P7 began by reading the README and spending an extended period forming an overall understanding of the task. P7 skipped creating a virtual environment and spent most of the early task time
P8 began by inspecting the README before starting and set up the environment in a separate terminal, completing most steps correctly but encountering repeated issues during environment crea
P9 first inspected the README independently and then shared the project directory with Ask mode, instructing it to read the README and implement a solution accordingly. Ask mode generated a
P10 began by reading the README and creating the virtual environment. P10 informed the agent that the environment was already activated and that required packages such as matplotlib and nu
P11 began by asking Ask mode to inspect the project and subsequently requested identification of the required packages. Based on the guidance provided, P11 started installing dependencies but
P12 followed the README step by step and completed most setup actions correctly. P12 experienced difficulties activating the script and spent time listing directory contents to verify the environm

task_1_likert_q_1 ("I felt in control of the coding process.")_1_worst_5_best	task_1_likert_q_2 ("I trusted the correctness of the AI's output.")_1_worst_5_best
2	3
3	4
2	3
3	4
1	2
4	2
1	5
1	4
3	5
5	5
3	5
4	3

task_1_likert_q_3 ("The tool/mode matched my natural workflow.")_1_worst_5_best
5
5
5
5
3
4
3
1
3
5
4
4

task_1_likert_q_4 ("The tool/mode helped me stay focused and in flow")_1_worst_5_best
3
5
3
5
3
3
3
2
2
4
5
3

task_1_likert_q_5 ("The tool/mode helped me complete the task efficiently.")_1_worst_5_best	Task2_Mode	Task2_Start_time	Task2_End_time	Task2_Duration_min
5	Agentic/Agent	12:51:00	12:58:00	7
5	Vibe/Ask	13:24:00	13:39:00	15
4	Agentic/Agent	16:30:00	16:32:00	2
5	Vibe/Ask	14:14:00	14:29:00	15
4	Agentic/Agent	15:36:00	15:41:00	5
3	Vibe/Ask	21:25:00	21:35:00	10
5	Agentic/Agent	18:31:00	18:35:00	4,2
5	Vibe/Ask	20:25:00	20:33:00	8
4	Agentic/Agent	19:15:00	19:17:00	2
5	Vibe/Ask	19:12:00	19:14:00	2
5	Agentic/Agent	19:30:00	19:32:00	2
5	Vibe/Ask	10:22:00	10:33:00	11

Task2_Success_0_fail_1_passed	Task2_Prompts_or_AgentRuns	Task2_Interruptions
1	2	0
0	9	2
1	1	0
0	12	5
1	4	0
1	6	0
1	1	0
1	5	1
1	1	0
1	1	0
1	2	0
0	3	0

	Task2_Notes
P1 initially scanned the README independently and then referenced the project directory when instructing Agent mode about the required process. P1 asked the agent to complete the task autono	
P2 re-inspected the README and followed the setup steps sequentially, starting by checking terminal requirements. Despite the terminal indicating that pip and Python were already installed, P2 r	
P3provided the issue, the agent inspected the project files, created the virtual environment, installed pytest, and executed the tests successfully. The task was completed with passing results, reso	
P4 provided the task question into Ask mode, which responded with a high-level to-do list and examples. Instead of providing the Task 1b README, P4 shared a general repository description. P4	
P5 provided the README to Agent mode but modified its contents, explicitly clarifying which files should be used. P5 closely inspected the code throughout execution and followed a highly algorith	
P6 read the README and task description again with increased attention, carefully inspecting each line of code and focusing on understanding the data analysis logic. P6 examined the data files i	
P7 instructed Agent mode to resolve a bug at a specific location in the codebase. The agent proceeded with the implementation process but skipped the recommended practice of creating a virtua	
P8 again started by reading the README and followed a traditional, step-by-step approach. P8 attempted to activate the virtual environment but failed and proceeded to inspect solution.py instead	
P9 provided the file location to Agent mode and requested it to complete the task according to the README, including running the task and resolving any issues encountered. P9 enabled autonom	
P10 moved quickly due to familiarity from the previous task. Without engaging in extended setup or discussion, P10 copied the README, stated “this is the task,” provided the relevant code conte	
P11 instructed Agent mode to inspect the project structure and then independently reviewed the README. After this initial clarification, the agent completed the challenge with a single approval fro	
P12 again began by reading the README and was already familiar with the overall process. P12 recreated the virtual environment but initially encountered issues during creation before successfu	

task_2_likert_q_1 ("I felt in control of the coding process.")_1_worst_5_best	task_2_likert_q_2 ("I trusted the correctness of the AI's output.")_1_worst_5_best
3	2
2	4
4	5
1	2
1	2
1	1
1	4
4	4
3	5
5	5
2	5
4	3

task_2_likert_q_3 ("The tool/mode matched my natural workflow.")_1_worst_5_best	task_2_likert_q_4 ("The tool/mode helped me stay focused and in flow")_1_worst_5_best
4	4
4	3
3	3
1	1
1	2
3	2
1	1
4	4
3	3
5	5
5	2
4	4

task_2_likert_q_5 ("The tool/mode helped me complete the task efficiently.")_1_worst_5_best
5
4
5
1
5
1
5
5
4
5
5
5

Interview("How did your sense of control differ between the two modes, and which one matched y
P1 reported a stronger sense of control in Agent mode, stating that it better matched their natural workflow. Although the agent performed the task autonomously, P1 emphasized the ability to inter
P2 reported that Agent mode felt smoother and better aligned with their natural workflow. Ask mode primarily provided instructions, which P2 experienced as slower and more fragmented, whereas
P3reported feeling somewhat lost in Ask mode. Although Agent mode involved less direct control, it better matched Ugur's natural workflow and reduced cognitive overhead related to setup and de
P4 reported that Agent mode better matched their natural workflow and provided a stronger sense of control. P4 attributed this to the agent's awareness of file locations and its ability to act with bro
P5 reported a stronger sense of control in Ask (Vibe) mode, noting that although they did not write the code themselves, they were able to follow the process closely and understand each step. In c
P6 reported a stronger sense of control in Agent mode, emphasizing the ability to review, accept, or reject changes while the agent handled execution and debugging. In contrast, Ask mode was p
P7 reported a stronger sense of control in Ask (Vibe) mode, as it aligned closely with their usual way of working and allowed them to inspect changes before committing them. Agent mode was als
P8 reported that Ask (Vibe) mode best matched their natural workflow. P8 preferred maintaining fine-grained control over file-specific changes and described Agent mode as more of an approval-c
P9 reported that Agent mode better matched their natural workflow and provided a stronger sense of control. P9 emphasized that Agent mode could autonomously handle iterations and refinemen
P10 reported feeling more in control when using Ask (Vibe) mode, describing it as better aligned with a human-in-the-loop workflow. Agent mode was perceived as too autonomous, with P10 main
P11 reported that Agent mode better matched their natural workflow. Although Ask mode provided explicit step-by-step control, P11 did not perceive a loss of control in Agent mode, emphasizing t
P12 reported a stronger sense of control in Ask (Vibe) mode, as it more closely resembled their usual way of coding and allowed them to remain actively involved in each step. Agent mode was pe

Interview("Which mode gave you more confidence in the correctness of the results, a
P1 expressed higher confidence in the correctness of results produced by Agent mode, describing the outputs as clean and unexpectedly optimal.
P2 expressed greater confidence in the results produced by Agent mode, highlighting that the agent also conducted testing, which increased trust in the correctness of the output.
P3expressed greater confidence in the correctness of results produced in Agent mode, as it allowed them to focus on analyzing the task output rather than spending time on configuration issues.
P4 expressed higher confidence in Agent mode outputs, noting that the agent could debug issues autonomously, allowing P4 to focus on prompting rather than file navigation.
P5 expressed greater confidence in the results produced in Ask mode, emphasizing that they understood the logic and reviewed the function before running tests. P5 stated that, for production use
P6 expressed higher confidence in Agent mode results, stating that Ask mode outputs were incorrect, whereas Agent mode produced more reliable solutions.
P7’s confidence in correctness was primarily tied to the presence of test cases. While Agent mode increased trust when tests were visible, P7 expressed reservations about relying on it without tho
P8 indicated similar levels of correctness across both modes but expressed greater confidence when using Ask mode. The ability to validate precise, incremental changes improved focus and mad
P9 expressed greater confidence in the correctness of results produced by Agent mode, noting that iterative loops and error detection were more transparent and easier to notice.
P10 expressed higher trust in Vibe Coding. Although both modes ultimately passed the tests, P10 valued having visibility into the reasoning and intermediate steps. For P10, Agent mode felt reliab
P11 expressed higher confidence in the correctness of results produced by Agent mode, stating that the agent’s outputs were superior and, in their view, more reliable than those of a human deve
P12 expressed higher confidence in Ask mode outputs, noting that receiving smaller, understandable code snippets enabled direct review and selective use. Although Agent mode provided some

Interview("In which mode did you feel more efficient and able to stay in flow during the task?")
P1 felt more efficient in Agent mode and reported better flow, despite the agent completing the task independently. P1 noted that Agent mode outperformed a human developer in efficiency and co
P2 felt more efficient and able to maintain flow in Agent mode. The agent’s ability to provide code, manage context such as file locations, and support implementation contributed to a more stream
P3felt more efficient and less distracted in Agent mode. In Ask mode, frequent context switching occurred, including the urge to search externally, whereas Agent mode supported sustained focus
P4 felt more efficient and less interrupted in Agent mode. The autonomous handling of setup, navigation, and execution supported sustained flow throughout the task.
P5 perceived Agent mode as more efficient in terms of task completion but reported lower flow due to increased distraction. Ask mode provided P5 with a greater sense of control, which supporte
P6 felt more efficient and able to stay in flow in Agent mode. The step-by-step execution and interactive feedback supported smoother progress and sustained engagement.
P7 felt more efficient in Agent mode but did not experience strong flow. Despite faster progress, the reduced sense of control limited sustained engagement.
P8 perceived Agent mode as more efficient in terms of speed, as changes were applied quickly with minimal input. However, Ask mode provided better flow, as active involvement in each step su
P9 did not report strong flow in either mode. Due to time constraints, P9 skipped explicit planning and relied on a brute-force approach, which limited perceived efficiency and sustained flow across
While acknowledging that Agent mode was faster, P10 reported better flow in Vibe Coding. Seeing the reasoning and progression supported sustained focus, whereas Agent mode completed the
P11 felt more comfortable and efficient when using Agent mode. The reduction in manual effort supported sustained flow and a smoother task experience.
P12 acknowledged that Agent mode was faster in terms of raw efficiency. However, for long-term use and maintainability, Ask mode was preferred due to better code comprehension, even if it req

Video_Record_1_exist_0_doesnotexist
1
1
1
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